Functions And Cross-Sections

Distribution Functions

The coulomb cross-section is much greater than the D-T fusion cross-section. This tells us that the random coulomb collisions will lead to thermodynamic equilibrium on a time scale much shorter than fusion collision time.

Thermodynamic equilibrium (from statistical mechanics) is described by the maxwellian distribution function:

$$f_j(\vec{r},\vec{v},t) = n_j \left(\frac{m_j}{2\pi T_j}\right)^{\frac{3}{2}} e^{-\frac{m_j v^2}{2T_j}} \rightarrow v^2 = v_x^2 + v_y^2 + v_z^2$$

n_j → Number density

T_j → Temperature

j denotes D or T

Fusion Cross-section and Power Density

The starting point for our cross-section will be the hard-sphere. Here the cross section will be given by:

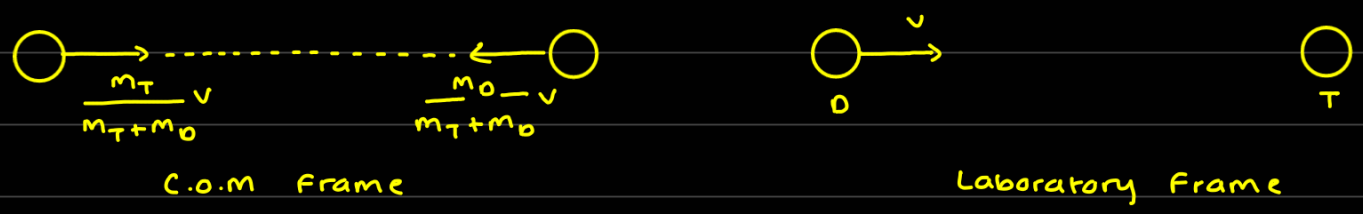
$$\sigma = \pi d^2 \approx 0.8 \times 10^{-28}\, m^2 \approx 1 \text{ barn} \qquad (1 \text{ barn} = 10^{-28}\, m^2)$$

d → Nuclear diameter

This cross-section is very crude and fails to account for the velocity dependence caused by the coulomb force.

To solve this, we can try out a classical model:

This model is quite simple, it basically states that the kinetic energy of D and T combined must be greater than the coulomb potential energy.



Mathematically, a collision occurs only if:

$$\frac{m_D}{2}\left(\frac{m_T}{m_D+m_T} v\right)^2 + \frac{m_T}{2}\left(\frac{m_D}{m_D+m_T} v\right)^2 \geq \frac{e^2}{4\pi\epsilon_0 d}$$

In other terms (subbing the value of d):

$K_{cm} \geq 288\, keV$

K_{cm} → kinetic energy of the system in C.o.M frame

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This result is very hard to achieve, 288 keV being a hard barrier on the initiation of the reaction.

Luckily for us, this model does not account for Quantum Mechanics.